

W2-Ex2 — Playing with the Embeddings Explorer

Demo `week-02/demos/03-embeddings-explorer`, running `nomi-embed-text` on local Ollama. Every sentence comes back as 768 numbers with `norm = 1.0000` — a point on the unit sphere. “Similar” then just means “near”, and you can put a number on it.

1 — MEANING, NOT WORDS

text: how quick is the warehouse robot?

Closeness to the texts you entered before

#	cosine ↑	distance ↓	earlier text
1	0.865	0.519	The warehouse robot's top speed is 2.4 metres per second.
2	0.778	0.667	How fast does the robot go?
3	0.470	1.030	Acme Robotics was founded in 2019 in Girona.
4	0.331	1.156	The recipe needs two eggs and a cup of flour.

nearest first · cosine high = close, distance low = close — same order, because the vectors are unit length

After `/seed`, the query “how quick is the warehouse robot?” compared against the four seeded sentences.

The winner at **0.865** is “The warehouse robot's top speed is 2.4 metres per second.” — and the only words the two share are *the*, *warehouse* and *robot*. It does not contain “quick”, and my query does not contain “speed” or “fast”. The cake recipe falls to the bottom at **0.331**, below even the Acme Robotics line, which is about robots but not about speed. The ranking is tracking what the sentence is *about*, not which words it reuses. **This is the case that matched what I expected.**

2 — TWO NAMES, ONE GEOMETRY

Because the model returns unit-length vectors, the two measures are not independent: they are locked together by $\text{distance}^2 = 2 \cdot (1 - \text{cosine})$, so the distance is a strictly decreasing function of the cosine and whichever sentence has the highest cosine must have the lowest distance. There is only one quantity here — the angle between the two points. Cosine reports it directly; Euclidean distance reports the length of the chord that the angle subtends. Checking the identity against the numbers the demo printed:

```
cos 0.865 → √(2·(1-0.865)) = 0.520 ≈ 0.519 shown
cos 0.331 → √(2·(1-0.331)) = 1.157 ≈ 1.156 shown
cos 0.955 → √(2·(1-0.955)) = 0.300 = 0.300 shown
```

3 — TRYING TO FOOL IT

text: The robot is not fast.

The embedding — meaning as coordinates

`nomi-embed-text` · 768 dimensions · norm = 1.0000 (unit length)
 [+0.0287, +0.0754, -0.1580, +0.0118, +0.0678, +0.0819, +0.0227, +0.0117, -0.0330, -0.0336, +0.0026, +0.0488, ...] (first 12 of 768)

Closeness to the texts you entered before

#	cosine ↑	distance ↓	earlier text
1	0.955	0.300	The robot is fast.

nearest first · cosine high = close, distance low = close — same order, because the vectors are unit length

The surprise. “The robot is not fast.” against “The robot is fast.” — cosine **0.955**, distance 0.300.

text: Me gusta mucho esta película

The embedding — meaning as coordinates

`nomi-embed-text` · 768 dimensions · norm = 1.0000 (unit length)
 [+0.0689, +0.0465, -0.1253, +0.0143, -0.0470, -0.0205, -0.0641, -0.0177, -0.0161, +0.0328, +0.0296, +0.0451, ...] (first 12 of 768)

Closeness to the texts you entered before

#	cosine ↑	distance ↓	earlier text
1	0.776	0.670	M'encanta aquesta pel·lícula
2	0.473	1.026	I love this film
3	0.405	1.091	The recipe needs two eggs and a cup

nearest first · cosine high = close, distance low = close — same order, because the vectors are unit length

The second surprise. One meaning in three languages. Catalan and Spanish sit at **0.776**; the English version of the same sentence sits at **0.473**, barely above an unrelated cake recipe at 0.405.

The negation was the case that surprised me. Adding the single word *not* reverses the truth of the sentence and moves the point almost nowhere — 0.955 is nearer than any genuine paraphrase I produced in task 1. That tells me an embedding captures **topic** — what a sentence is about, which words it lives among — and largely ignores **polarity**: “near” means “about the same thing”, not “says the same thing”. The language test points the same way. “Me gusta mucho esta película” lands 0.776 from its Catalan twin but only 0.473 from its English translation — a gap of 0.303 for a sentence that means exactly the same thing — while an unrelated cake recipe sits at 0.405, just 0.068 further out than the translation. So the vector encodes which language the sentence is written in nearly as strongly as what it says; `nomi-embed-text` was trained mostly on English, and it shows. For RAG the consequence is direct — a manual saying “the warranty does not cover water damage” is exactly what gets retrieved for “is water damage covered?”, and the retriever cannot tell me which of the two it found. Retrieval finds the right paragraph; it does not read it.